Example

- Consider the following equations of motion:

Dynamic equilibrium: $m\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}\ddot{r}_1\\\ddot{r}_2\end{bmatrix}+k\begin{bmatrix}3 & 1\\1 & 3\end{bmatrix}\begin{bmatrix}r_1\\r_2\end{bmatrix}=\begin{bmatrix}R_1\\R_2\end{bmatrix}$

Free vibration: $\omega_1=\sqrt{2\frac{k}{m}},\quad \omega_2=\sqrt{4\frac{k}{m}}$

$$\Phi_1=\begin{bmatrix}1\\-1\end{bmatrix},\quad \Phi_2=\begin{bmatrix}1\\1\end{bmatrix}$$

In modal space:

The generalised coordinates q, such that $\mathrm{r}=\Phi\mathrm{q}$

$$\mathrm{m}\ddot{\mathrm{q}}+\mathrm{k}\mathrm{q}=\mathrm{Q}\quad\text{where}\begin{cases}\mathrm{m}=\Phi^t\mathrm{M}\Phi\\ \mathrm{k}=\Phi^t\mathrm{K}\Phi\quad\text{and}\quad\Phi=[\Phi_1\ \ \Phi_2]\\ \mathrm{Q}=\Phi^t\mathrm{R}\end{cases}$$

1. Assuming that $\begin{bmatrix}R_1\\R_2\end{bmatrix}=\begin{bmatrix}R_0\\R_0\end{bmatrix}\cos(\omega t)$ where R_0 is a constant, obtain the dynamic steady state response.

   50%

2. No loads are applied to DOF 2, and a constant load of magnitude R_0 is applied to DOF 1 for time $t>0$. Calculate the response $r_1(t)$, assuming that the mass is at rest at its equilibrium position at time $t=0$.

   50%

1. $\begin{bmatrix}r_1\\r_2\end{bmatrix}=\begin{bmatrix}r_0\\r_0\end{bmatrix}\cos(\omega t)$ $\quad\therefore\begin{bmatrix}\ddot{r}_1\\\ddot{r}_2\end{bmatrix}=-\omega^2\begin{bmatrix}r_0\\r_0\end{bmatrix}\cos(\omega t)=-\omega^2\begin{bmatrix}r_1\\r_2\end{bmatrix}$

$$\therefore\begin{bmatrix}-\omega^2m+3k & k\\k & -\omega^2m+3k\end{bmatrix}\begin{bmatrix}r_1\\r_2\end{bmatrix}=\begin{bmatrix}R_1\\R_2\end{bmatrix}$$

$$\therefore\begin{bmatrix}r_1\\r_2\end{bmatrix}=\frac{1}{\Delta}\begin{bmatrix}-\omega^2m+3k & -k\\-k & -\omega^2m+3k\end{bmatrix}\begin{bmatrix}R_1\\R_2\end{bmatrix}$$

where $\Delta=(-\omega^2m+3k)^2-k^2$

$$\therefore r_0=\frac{1}{\Delta}\begin{bmatrix}-\omega^2m+3k & -k\\-k & -\omega^2m+3k\end{bmatrix}\begin{bmatrix}R_0\\R_0\end{bmatrix}$$

$$=\frac{1}{\Delta}\begin{bmatrix}-\omega^2m+2k\\-\omega^2m+2k\end{bmatrix}R_0$$

2. $$m = \Phi^T M \Phi = \begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix} m \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} = m\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} = m\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}$$

$$k = \Phi^T K \Phi = \begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix} k \begin{bmatrix}3 & 1\\ 1 & 3\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} = k\begin{bmatrix}2 & -2\\ 4 & 4\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} = k\begin{bmatrix}4 & 0\\ 0 & 8\end{bmatrix}$$

$$Q = \Phi^T R = \begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}R_1\\ R_2\end{bmatrix} = \begin{bmatrix}R_1 - R_2\\ R_1 + R_2\end{bmatrix} = \begin{bmatrix}0\\ 2\end{bmatrix} R_0 \cos(\omega t)$$

$$m\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}\ddot{q} + k\begin{bmatrix}4 & 0\\ 0 & 8\end{bmatrix} q = \begin{bmatrix}R_1 - R_2\\ R_1 + R_2\end{bmatrix} = \begin{bmatrix}0\\ 2\end{bmatrix} R_0 \cos(\omega t)$$

$$q = \int_0^t \frac{1}{\beta m} e^{-\alpha(t-\tau)} \sin(\beta(t-\tau)) R(\tau)\, d\tau$$

$$= \int_0^t \frac{1}{\omega \underset{2m}{m}} \sin(\omega(t-\tau)) R_0\, d\tau$$

$$= \frac{R_0}{\omega \underset{2m}{m}} \left[\frac{1}{\omega}\cos(\omega(t-\tau))\right]_0^t$$

$$= \frac{R_0}{\omega \underset{2m}{m}}\left(\frac{1}{\omega} - \frac{1}{\omega}\cos(\omega t)\right) = \frac{R_0}{4k}(1 - \cos(\omega_1 t)) \quad \text{for DoF 1, } \omega_1 = \sqrt{\frac{2m}{k}}$$

$$\text{or} = \frac{R_0}{8k}(1 - \cos(\omega_2 t)) \quad \text{for DoF 2, } \omega_2 = \sqrt{\frac{4m}{k}}$$

$$\therefore q = \begin{bmatrix}q_1\\ q_2\end{bmatrix} = \frac{R_0}{4k}\begin{bmatrix}1 - \cos(\omega_1 t)\\ 2 - 2\cos(\omega_2 t)\end{bmatrix}$$

$$\therefore r = \Phi q = \begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} q = \frac{R_0}{4k}\begin{bmatrix}3 - 2\cos(\omega_2 t) - \cos(\omega_1 t)\\ 1 + \cos(\omega_1 t) - 2\cos(\omega_2 t)\end{bmatrix}$$